ESERCIZIO POL1 PMOS

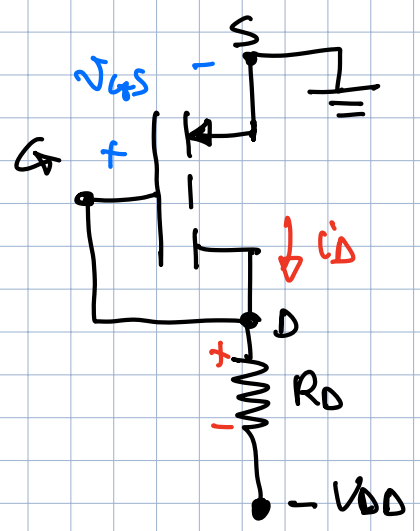
CALCOLARE i_D DI PMOSFET SAPENDO CHE

$W = 10\mu m$, $L = 0{,}5\,\mu m$, $K_p' = 40\,\mu A/V^2$ $V_{TP} = -0{,}75V$

a) $v_{GS} = -0{,}5V$, $v_{DS} = -0{,}2V$
b) $v_{GS} = -1{,}1V$, $v_{DS} = -0{,}2V$
c) $v_{GS} = -1{,}1V$, $v_{DS} = -0{,}8V$
d) $v_{GS} = -1{,}4V$, $v_{DS} = -0{,}8V$

[0, 40 μA, 49 μA, 169 μA]

ESERCIZIO POL2 PMOS



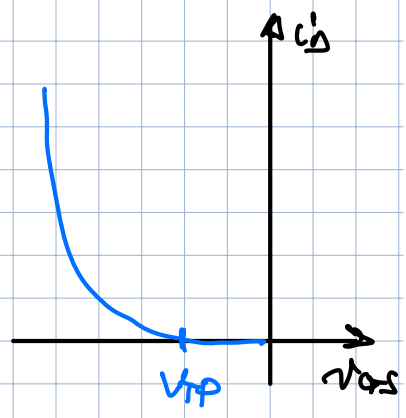
$V_{DD} = 15V$

$R_D = 75\,K\Omega$

$V_{TP} = -0{,}75\,V$

$K_p' = 10\,\mu A/V^2$

$\frac{W}{L} = 10$



CALCOLARE i_D, v_{DS}

[$i_D = 165{,}35\,\mu A$, $v_{DS} = -2{,}6V$]

ESERCIZIO POL3 PMOS

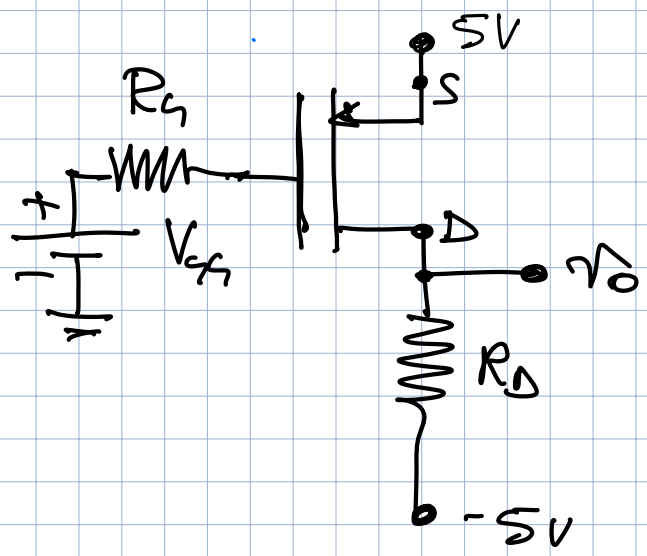
$V_G = 3V$

$R_D = 1\,K\Omega$

$R_G = 100\,K\Omega$

$K_p = 0{,}5\,mA/V^2$

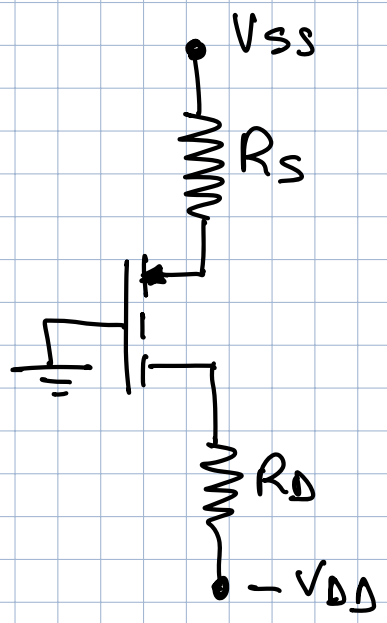
$V_{TP} = 2V$



CALCOLARE i_D, v_{DS}

[$i_D = 4mA$, $v_{DS} = -6V$]

ESERCIZIO POL4 PMOS



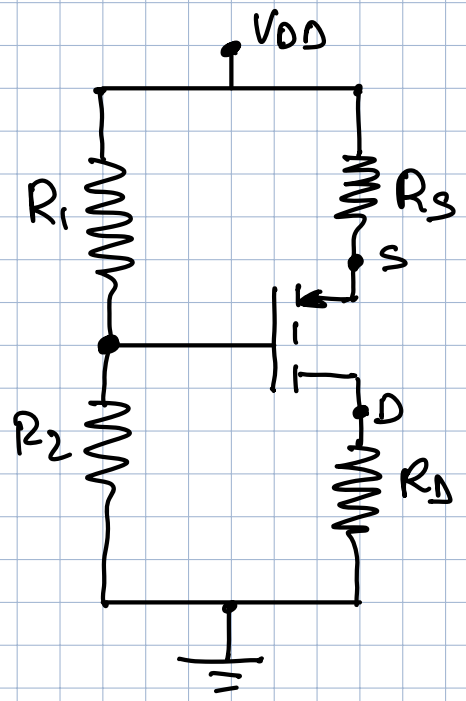
$V_{DD} = 15V$ $V_{SS} = 15V$

$K_P = 2\,mA/V^2$ $V_{TP} = -3V$

Determinare R_D e R_S in modo che $I_D = 1\,mA$, $V_{DS} = -3V$

[$R_S = 11\,K\Omega$; $R_D = 16\,K\Omega$]

ESERCIZIO POL5 PMOS



DATI: $V_{DD} = 15V$

$R_1 = 200\,K\Omega$ $R_2 = 400\,K\Omega$

$R_S = 0{,}25\,K\Omega$ $R_D = 1{,}0\,K\Omega$

$K_P = 4\,mA/V^2$

$V_{TP} = -1V$ $\lambda_P = 0\,V^{-1}$

CALCOLARE I_D, V_{DS}, V_{GS}

[$V_{GS} = -3V$, $I_D = 8\,mA$, $V_{DS} = -5V$]

ESERCIZIO POL5B PMOS

USANDO LO SCHEMA SOPRA (POL5PMOS):

$V_{DD} = 15V$ $R_1 = 200\,K\Omega$

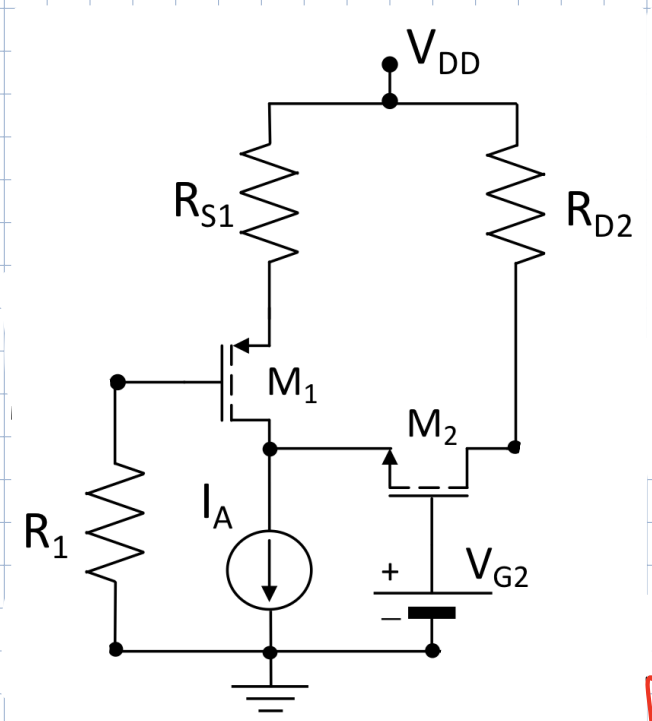
$R_D = 1{,}5\,K\Omega$ $R_2 = 200\,K\Omega$

$K_P = 10\,mA$, $V_{TP} = -1V$ $\lambda_P = 0$

TROVARE R_S CHE GARANTISCA $I_{D1} = 5\,mA$, e CALCOLARE V_{GS} e V_{DS}

[$R_S = 1{,}1\,K\Omega$, $V_{GS} = -2V$, $V_{DS} = -2V$]

ESERCIZIO POL6 PMOS



Dati:
V_{DD}=12 V,
V_{G2}=4 V,
I_A=16 mA

R_1 = 500 kΩ,
R_{S1} =1 kΩ,
R_{D2} =1 kΩ,

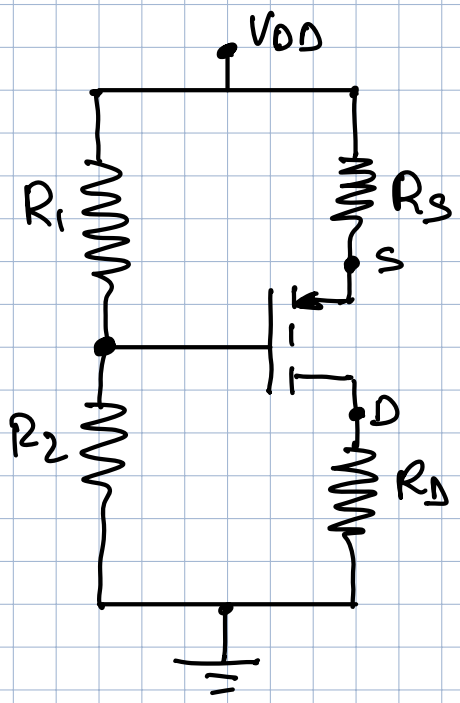
M_1: k_{p1}=4 mA/V^2,
V_{TP1} = -2 V,
λ_{p1}=0 V^{-1};

M_3: k_{n2}=4 mA/V^2,
V_{TN2} = 2 V,
λ_{n2} = 0 V^{-1}.

CALCOLARE I_{D_1}, V_{DS_1}, V_{GS_1}
I_{D_2}, V_{DS_2}, V_{GS_2}

[$V_{GS_1} = -4V$, $I_{D_1} = 8mA$, $V_{DS_1} = -4V$]
[$V_{GS_2} = 4V$, $I_{D_2} = 8mA$, $V_{DS_2} = 4V$]

ESERCIZIO POL7 PMOS



$V_{DD} = 10V$
$R_1 = 1 k\Omega$,
$R_2 = 1 k\Omega$
$R_S = 39 k\Omega$
$R_D = 75 k\Omega$
$K_p = 25 \mu A/V^2$
$V_{TP} = -1V$

TROVARE IL PUNTO DI LAVORO
V_{GS}, I_D, V_{DS}

[$V_{GS} = -2,66V$
$I_D = 34,45 \mu A$
$V_{DS} = -6,07V$]